Lab 5 - 2D Isometric Drawing

Course: WSQ Technical Drawing with AutoCAD (TGS-2021005538) | Topic 02: Technical Drawing with AutoCAD

Objective: LO2 - produce 2D and 3D technical drawings using AutoCAD (A2, K2)

Goal

Use ISODRAFT to create a 2D isometric representation of the 3D object supplied in iso-object.dwg, switching isoplanes and drawing isocircles for the round features.

You'll produce

A flat isometric drawing of the 3D object - edges on the three isoplanes plus isocircle holes.

Commands / tools: ISODRAFT, isoplanes (F5), ELLIPSE Isocircle, polar tracking

Files in this lab

- iso-object.dwg - AutoCAD drawing file
- iso-object-sheet.pdf - drawing sheet / reference (PDF)
- iso-object.png - reference image

Step-by-step

1. Open the object sheet iso-object-sheet.pdf and identify its three faces
2. Turn on isometric drafting from the status bar and set the Top isoplane

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ISODRAFT → Isoplane Top (F5)
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3. Draw the top face edges along the 30°/150° isometric axes with polar tracking

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L with polar 30°/150°
```

4. Switch to the Right and Left isoplanes and complete the vertical faces

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F5 → Right | F5 → Left
```

5. Add the round features as isocircles on the correct isoplane

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EL → I (Isocircle) → centre → radius
```

6. Compare your isometric with the 3D model in iso-object.dwg and save

```
SAVEAS iso-mine.dwg
```

Check your work

Each face lies on the correct isoplane (edges at 30°/90°/150°), the isocircles read as ellipses aligned to their faces, and the drawing matches the supplied object's proportions.